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Illumination-aware softmax guided shadow removal[☆]Lianmeng Wei[✉], Sihui Luo^{✉*}

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ABSTRACT

While recent learning-based methods have boosted the performance of shadow removal, a major challenge persists: most leading approaches rely on manually annotated ground-truth masks as auxiliary priors. However, acquiring such manual annotations is costly, and model performance often degrades sharply without ground-truth mask guidance. To tackle this problem, we propose a multi-scale illumination-aware softmax generation method. Specifically, we compute the luminance ratio between the shadow image and its shadow-free counterpart, followed by multi-scale filtering and fusion to produce a coherent softmax. This softmax is learned and predicted via a shallow network, which subsequently guides the restoration process. Compared to binary ground-truth masks, our approach yields softmax with improved coherence and more accurate preservation of edge gradients. Furthermore, we introduce a synergistic fusion of structural feature derived from self-extracted multi-scale representations using Gaussian kernels, which effectively retains structural information within shadowed regions. Extensive experiments confirm that our model achieves state-of-the-art performance. Our code is available at <https://github.com/welemon7/Softmask>.

1. Introduction

Shadow removal, which is dedicated to correcting illumination discrepancies and recovering structural texture in shadow regions, serves as a vital preprocessing step for a range of applications such as UAV remote sensing [1], autonomous driving systems [2], and visual servoing [3].

Recent years have seen significant progress in shadow removal driven by deep learning. Methods such as ShadowFormer [4], which aggregates global context via channel attention to reduce computational complexity, and HomoFormer [5], which employs shuffle operations to achieve spatially uniform shadow distribution, exemplify this advancement. Despite their impressive performance, these mainstream methods uniformly depend on manually annotated ground-truth masks as auxiliary priors to guide the reconstruction. As shown in Fig. 1, when HomoFormer [5] is not supervised by the ground-truth (GM) prior, its performance degrades markedly.

Acquiring such manual annotations, however, is costly. Consequently, recent research has shifted its focus toward shadow removal that does not require ground-truth masks. These include DeS3 [7] combines diffusion models with adaptive mechanisms, and OmniSR [8] integrates DINO features [9] with semantic cues [10] to enhance performance. Nevertheless, such GM-free methods often struggle with complex and varying illumination, while also incurring high computational costs.

To address the challenge of shadow removal without ground-truth masks, we propose an innovative softmax prediction method that eliminates reliance on manual annotations while providing physically more accurate and structure-preserving guidance for removal. Following physical illumination principles, our softmax is generated in linear color space by computing precise luminance ratios between shadow and shadow-free image pairs. Multi-scale illumination modeling is further introduced to capture shadow characteristics across different scales. As shown in Fig. 2, our softmax generation method accurately localize shadow regions with more coherent boundaries than manual annotations, offering strong and reliable guidance for learning the softmax-prediction mapping. Additionally, multi-scale structural information-such as edges and textures extracted via Gaussian filtering-is incorporated to offer essential structural cues for the removal network. Our shadow removal follows a two-stage pipeline: Network I is trained to predict the softmax from the shadowed input; Network II then takes both the shadow image and the predicted softmax to reconstruct the shadow-free result.

In summary, our main contributions are two-fold: We propose a novel softmax generation method that produces high-quality, coherent softmaxes from only shadow and shadow-free image pairs, offering more physically plausible estimates compared to conventional binary ground-truth masks. We design a structural extractor that captures rich

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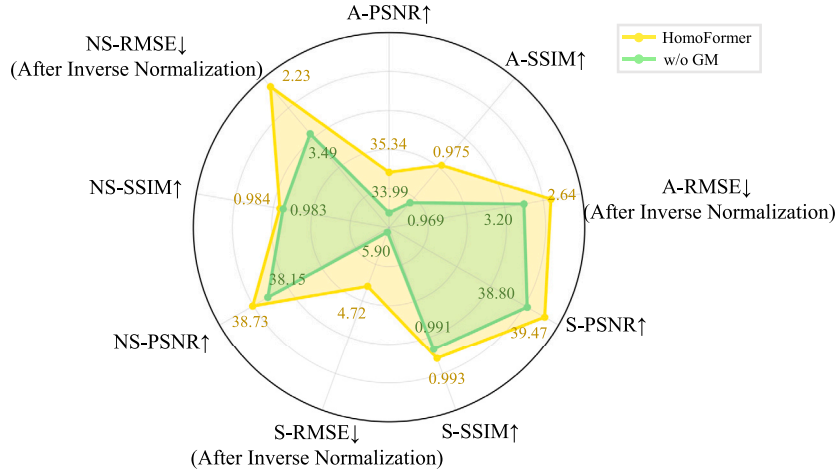


Fig. 1. HomoFormer [5] vs. HomoFormer (w/o GM) on ISTD+ dataset [6]. PSNR/SSIM: higher is better (positive metric), RMSE: lower is better (negative metrics shown with inverse normalization).

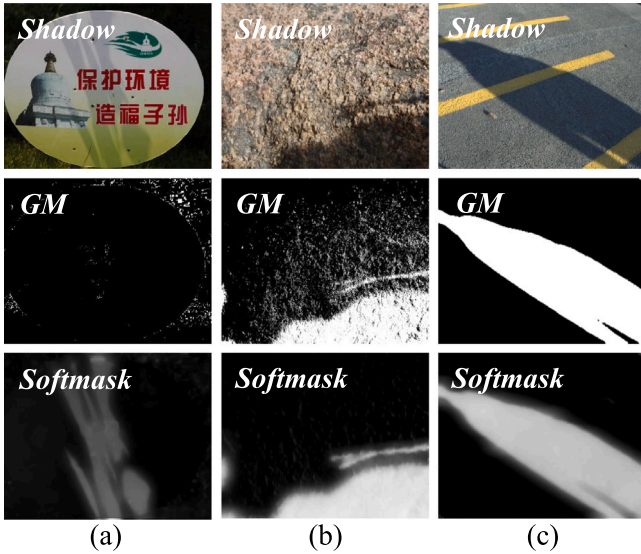


Fig. 2. The ground-truth masks (GM) and our softmask for three samples. Compared to the ground-truth masks, our softmask eliminates annotation inaccuracies and inconsistent boundaries.

multi-scale structural information, effectively enhancing the performance of baseline models. Extensive experiments demonstrate that our approach significantly surpasses baseline and achieves state-of-the-art results on multiple benchmark datasets.

2. Related work

Shadows present significant challenges for image restoration due to their diverse shapes, structures, and coverage. Deep learning has driven remarkable progress in this area. Some approaches rely on manually annotated ground-truth shadow masks to guide image restoration, such as RASM [11], which utilizes region-aware attention to aggregate features from both shadow and non-shadow areas, and FW-Former [12], features a region-aware attention mechanism for better restoration of fine details by focusing on information around shadow boundaries. Recently, diffusion models have also been explored for shadow removal. ShadowDiffusion [13] leverages generative priors for natural shadow-free results, and Diff-Shadow [14] further incorporates re-weighted cross-attention and global-guided sampling to maintain illumination consistency.

The rapid advancement of deep learning has led to a proliferation of data-driven shadow removal methods. The current paradigm is dominated by U-shaped networks, often integrated with Transformer modules to capture long-range dependencies, which have significantly boosted performance, robustness, and generalization. Representative works include ShadowFormer [4], which enhances context correlation via shadow-interaction attention; OmniSR [8], which enriches semantic cues using DINO features and geometric priors; and Soft-Shadow [15], leverages the Segment Anything Model and physical illumination constraints to generate softmasks for shadow removal. While these approaches push performance boundaries, the recent rise of diffusion models offers a new frontier. Methods like ShadowDiffusion [13] and DeS3 [7] treat removal as an iterative denoising process, achieving impressive shadow-free image generation.

Subsequently, the high cost of ground-truth mask annotation and the severe performance drop without them have motivated GM-free methods, DC-ShadowNet [16] inspired this direction by using a domain classifier, though with limited success. Building on this, DeS3 [7] employed diffusion models but remains constrained by shadow location and high parameters. Other works like ShadowRefiner [17] and OmniSR [8] incorporate frequency or other priors, yet still face performance bottlenecks.

Unlike these existing methods, we introduce a novel illumination-aware softmask generation approach and enable the model to learn the mapping from the shadow image to this softmask, which is then integrated into the main shadow removal network to eliminate the need for costly manual annotations and improve the performance of methods that operate without ground-truth masks.

3. Methodology

3.1. Overview

Based solely on paired shadow images I_s and shadow-free images I_{gt} without any manually annotated ground-truth masks, our framework (shown in Fig. 3) is built on a U-shaped Transformer architecture [18] that comprises two networks: (I) SoftMask Predictor – a shallow network that takes the shadow image I_s as input and outputs predicted softmask $\hat{M}_s$, used to guide the subsequent shadow removal stage. The supervisory signal M_s for softmask is derived from the multi-scale illumination-aware contrast between I_s and I_{gt} , as described in Section 3.2. (II) Shadow Removal – receives the shadow image I_s along with the softmask $\hat{M}_s$ predicted by Network I. It integrates a Multi-granularity Local Feature Enhancement (MLFE) module at each encoder-decoder layer to enrich and complement texture representation, ultimately producing the predicted shadow-free image I_d .

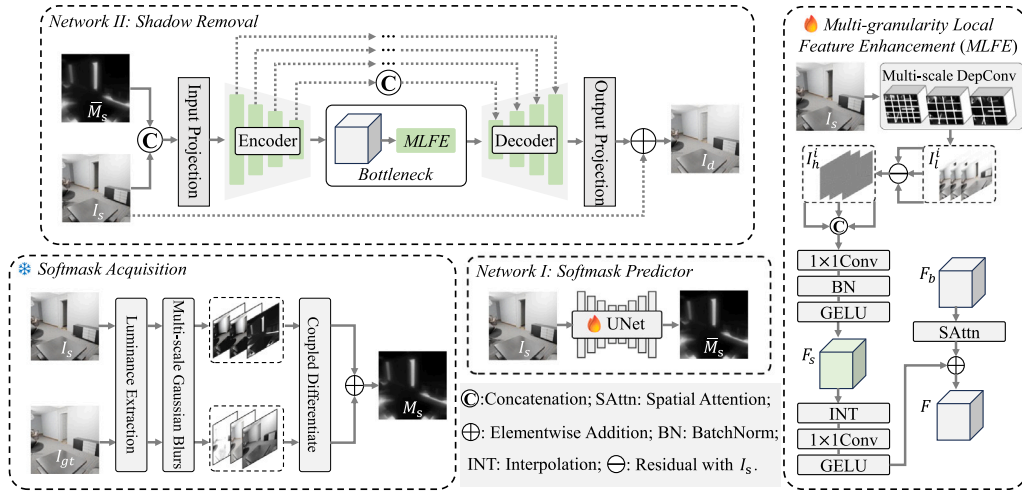


Fig. 3. The top panel illustrates the universal framework for applying the softmask beyond the removal network. The bottom panel shows the softmask predictor, which is trained with softmask supervision. The right panel depicts the MLFE module, used to enhance structural representations within the removal network.

3.2. Softmask acquisition

3.2.1. Illumination map acquisition

To approximate shadow regions, we first analyze the illumination discrepancy between the shadow image I_s and its shadow-free counterpart I_{gt} . In physical imaging, light intensity relates linearly to scene brightness. However, standard RGB images are typically gamma-encoded, introducing a nonlinear mapping between pixel values and actual radiance. Therefore, we convert both I_s and I_{gt} into linear RGB space [19] to adhere to a physically based lighting model prior to illumination comparison. Specifically, following the sRGB standard transformation defined in IEC 61966-2-1 [20], each channel is linearized as:

$$I_L = \begin{cases} \frac{I}{12.92} & I \leq 0.04045 \\ \left(\frac{I+0.055}{1.055}\right)^{2.4} & I > 0.04045 \end{cases} \quad (1)$$

where $I \in [0, 1]$ denotes the gamma-encoded pixel value and I_L is the corresponding linear radiance value.

We then compute the luminance maps L_s and L_{gt} using the standard photometric weighting coefficients that reflect human spectral sensitivity [21]:

$$L = 0.299 \times R + 0.587 \times G + 0.114 \times B \quad (2)$$

3.2.2. Multiscale Gaussian blur

Real-world shadows exhibit smooth illumination transitions and vary in size and edge softness. To model these characteristics at multiple scales, we apply Gaussian filtering [22] with different kernel sizes to both L_s and L_{gt} , obtaining smoothed luminance maps $\hat{L}_s$ and $\hat{L}_{gt}$, respectively. For a Gaussian kernel i , we compute the smoothed illumination as follows:

$$\hat{L}^i = \mathcal{G}_i(L) \quad (3)$$

Larger kernels capture broader shadow regions with smoother boundaries but may lose fine details, whereas smaller kernels retain local shadow structures but are more sensitive to noise. We then integrate the discrepancy illuminance maps $\hat{L}_s^i$ and $\hat{L}_{gt}^i$ of multiple scales via summation followed by a sigmoid activation that serves to squash the aggregated responses into a bounded range and introduce a smooth transition, yielding the final softmask M_s :

$$M_s = \text{Sigmoid} \left(\sum_{i=1}^n \left(1 - \frac{\hat{L}_s^i}{\hat{L}_{gt}^i} \right) \right) \quad (4)$$

where $n = 3$ denotes the number of Gaussian kernels used, with sizes set to 15, 35, and 75. This multi-scale integration enhances spatial continuity in illumination transitions and improves robustness to noise.

The acquired M_s for each training sample is then used to train the softmask predictor network.

3.3. Model architecture

3.3.1. Softmask predictor network

We adopt a Transformer-based U-Net architecture [18], which consists of an encoder, a decoder, and a bottleneck layer, to predict the softmask. Given a shadow image I_s , an initial feature is first projected from it and then passed through the encoder-decoder hierarchy. Each encoder, decoder, and bottleneck layer follows the same computational flow: the input feature x_{in} is normalized via LayerNorm, partitioned into non-overlapping windows for window-based multi-head self-attention $WAttn$, and subsequently restored to its original layout. The resulting feature is normalized again and processed by a multi-layer perceptron (MLP) block, which consists of a linear layer, a deconvolutional layer, a channel-split operation followed by elementwise multiplication for feature interaction, and a final linear layer. Another residual connection then yields the layer output x_{out} . This process can be summarized as:

$$x_1 = x + WAttn(\text{LayerNorm}(x_{in})) \quad (5)$$

$$x_{out} = x_1 + \text{MLP}(\text{LayerNorm}(x_1))$$

Downsampling is implemented by a 4×4 convolution with stride 2, doubling the channel dimension while halving the spatial size. Upsample uses a 2×2 transposed convolution with stride 2, halving the channel dimension while doubling the spatial size. Skip connections are added between corresponding encoder and decoder layers to preserve fine details. After the final decoder, an output projection layer yields the predicted softmask.

3.3.2. Shadow removal network

The shadow removal network retains the same core Transformer-U-Net structure as the softmask predictor, with the key modifications on the input/output setting and the integration of a structure-enhancing module.

The input is the channel-wise concatenation of the shadow image and the predicted softmask. After the final decoder layer, an output projection is applied, and its features are combined with the original shadow image via a residual connection to produce the shadow-free result.

Table 1

Quantitative comparisons with SOTA methods on ISTD+ dataset [6]. The best and the second best results are **boldfaced** and underlined, respectively. GM: Rely on manually annotated ground-truth shadow masks. “—” indicates unavailable data.

Method	GM	Shadow region			All image region		
		PSNR↑	SSIM↑	RMSE↓	PSNR↑	SSIM↑	RMSE↓
Input Image	—	20.88	0.945	37.93	20.51	0.908	8.31
AutoExposure (CVPR'21) [23]	✓	36.21	0.975	6.50	29.45	0.840	4.20
BMNet (CVPR'22) [24]	✓	37.30	0.990	6.19	32.30	0.955	3.60
ShadowFormer (AAAI'23) [4]	✓	39.67	<u>0.992</u>	5.20	35.46	0.973	2.80
ShadowDiffusion (CVPR'23) [13]	✓	39.82	—	4.90	35.68	0.970	2.70
HomoFormer (CVPR'24) [5]	✓	39.49	0.993	4.73	35.35	<u>0.975</u>	2.64
RASM (MM'24) [11]	✓	40.73	0.993	<u>4.41</u>	36.16	0.976	2.53
FW-Former (JVCI'25) [12]	✓	<u>40.11</u>	0.993	<u>4.71</u>	<u>35.84</u>	<u>0.975</u>	<u>2.58</u>
DC-ShadowNet (ICCV'21) [16]	✗	31.06	0.976	12.62	25.03	0.926	7.77
DeS3 (AAAI'24) [7]	✗	36.52	0.988	<u>6.62</u>	31.38	<u>0.958</u>	<u>3.94</u>
OmniSR (AAAI'25) [8]	✗	36.70	<u>0.992</u>	—	<u>33.34</u>	0.970	—
Ours	✗	39.20	0.993	5.61	34.29	0.970	3.29

Table 2

Quantitative comparisons with SOTA methods on SRD dataset [25]. The best and the second best results are **boldfaced** and underlined, respectively.

Method	GM	PSNR↑	RMSE↓
Input Image	—	18.19	14.05
AutoExposure(CVPR'21) [23]	✓	26.99	5.95
BMNet (CVPR'22) [24]	✓	31.69	4.46
ShadowFormer (AAAI'23) [4]	✓	32.90	4.04
ShadowDiffusion (CVPR'23) [13]	✓	34.73	3.63
HomoFormer (CVPR'24) [5]	✓	35.37	<u>3.33</u>
RASM (MM'24) [11]	✓	34.46	3.37
FW-Former (JVCI'25) [12]	✓	<u>34.88</u>	3.29
DC-ShadowNet (ICCV'21) [16]	✗	31.53	4.65
Refusion (CVPRW'23) [26]	✗	31.60	—
DeS3 (AAAI'24) [7]	✗	<u>34.11</u>	<u>3.72</u>
OmniSR (AAAI'25) [8]	✗	32.87	4.21
Detail-Preserving (CVPR'25) [27]	✗	33.63	—
Ours	✗	34.14	3.39

Table 3

Quantitative comparisons with SOTA methods on ISTD dataset [28]. The best and the second best results are **boldfaced** and underlined, respectively.

Method	GM	PSNR↑	SSIM↑
Input Image	—	20.56	0.893
DHAN (AAAI'20) [29]	✓	29.11	0.954
BMNet (CVPR'22) [24]	✓	30.28	0.959
ShadowFormer (AAAI'23) [4]	✓	32.21	0.968
ShadowDiffusion (CVPR'23) [13]	✓	32.33	0.969
FSR-Net (MM'23) [30]	✓	31.33	0.961
FW-Former (JVCI'25) [12]	✓	32.89	<u>0.971</u>
Diff-Shadow (AAAI'25) [14]	✓	33.69	0.976
DC-ShadowNet (ICCV'21) [16]	✗	26.38	<u>0.922</u>
Refusion (CVPRW'23) [26]	✗	25.13	0.871
ShadowRefiner (CVPRW'24) [17]	✗	28.75	0.916
OmniSR (AAAI'25) [8]	✗	30.45	0.964
Ours	✗	30.46	0.964

Besides, we integrate a lightweight structural extractor with the backbone to enhance texture representation. The shadow image I_s is smoothed by a set of depthwise convolutions with fixed multi-scale Gaussian kernels (kernel sizes $\{3, 5, 7\}$) to obtain low-frequency components I_l^i . High-frequency details I_h^i are then derived as the residual between the original input I_s and the smoothed versions I_l^i :

$$\begin{aligned} I_l^i &= \text{DepConv}(I_s) \\ I_h^i &= I_s - I_l^i \end{aligned} \quad (6)$$

Multi-scale high-frequency details are channel-wise concatenated, passed through a 1×1 convolution and a GELU activation [31], yielding the structural feature map F_s :

$$F_s = \text{GELU}(\text{Conv}(\text{Concat}(I_h^1, I_h^2, I_h^3))) \quad (7)$$

The structural feature F_s is integrated into each encoder-decoder layer through a fusion module. For a given base feature map F_b , spatial attention is first applied to enhance region-aware emphasis. Meanwhile, F_s is resized via interpolation, projected by a 1×1 convolution, BatchNorm and activated with GELU. The two are combined element-wise to obtain the final feature F :

$$F = \text{SAttn}(F_b) + \text{GELU}(\text{BN}(\text{Conv}(\text{INT}(F_s)))) \quad (8)$$

This design ensures that multi-scale structural cues are continuously injected throughout the network, improving texture recovery in shadow regions.

3.4. Loss function

We optimize the softmask predictor using the BCE loss [32]:

$$\mathcal{L}(\tilde{M}_s, M_s) = -[M_s \cdot \log(\tilde{M}_s) + (1 - M_s) \cdot \log(1 - \tilde{M}_s)] \quad (9)$$

Besides, Charbonnier loss [33] is utilized for learning the shadow removal model:

$$\mathcal{L}(I_d, I_{gt}) = \sqrt{\|I_d - I_{gt}\|^2 + \epsilon} \quad (10)$$

where $\tilde{M}_s, M_s$ represent the predicted softmask and the prior generated softmask. I_d, I_{gt} denote the prediction and ground-truth respectively, with $\epsilon = 10^{-6}$ ensuring numerical stability.

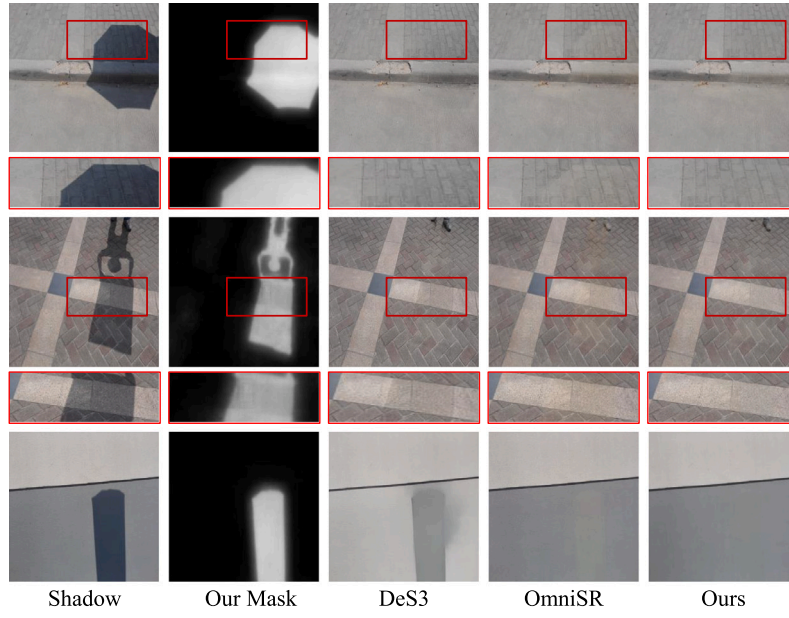
4. Results

4.1. Experimental setup

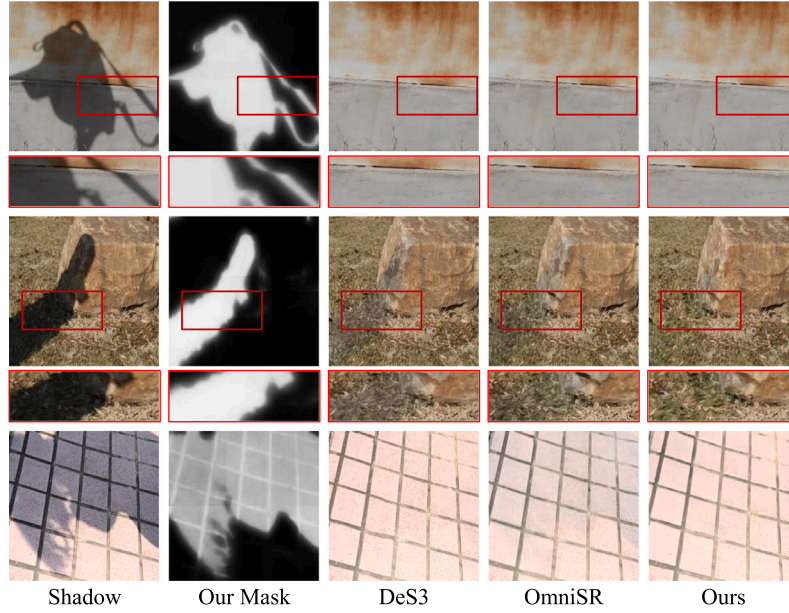
4.1.1. Datasets

We evaluate our model on five challenging public datasets. For evaluation metrics, the datasets used are:

- SRD Dataset [25]: This dataset contains 2680 image pairs for training and 408 for testing, each consisting of a shadow image and its shadow-free counterpart.
- ISTD Dataset [28]: It comprises 1870 triplets (shadow image, shadow-free image, and ground-truth mask), divided into 1330 training and 540 testing samples.
- Adjusted ISTD (ISTD+) Dataset [6]: An enhanced version of ISTD Dataset [28] that addresses illumination inconsistencies in the original data.



(a) Visual comparisons on ISTD+ dataset [6].



(b) Visual comparisons on SRD dataset [26].

Fig. 4. Visual comparisons of our approach with open-sourced SOTA methods.

- WRSD+ Dataset [34,35]: It Consists of 1000 training pairs and 100 test pairs, forming a high-resolution dataset without ground-truth masks.

For robustness evaluation, we adopt the LRSS dataset [36] and the USR dataset [37]. The LRSS dataset [36] comprises 46 paired shadow and shadow-free images, without providing ground-truth shadow masks. In contrast, the USR dataset [37] is an unpaired shadow removal benchmark, containing 2445 shadow images and 1770 shadow-free images.

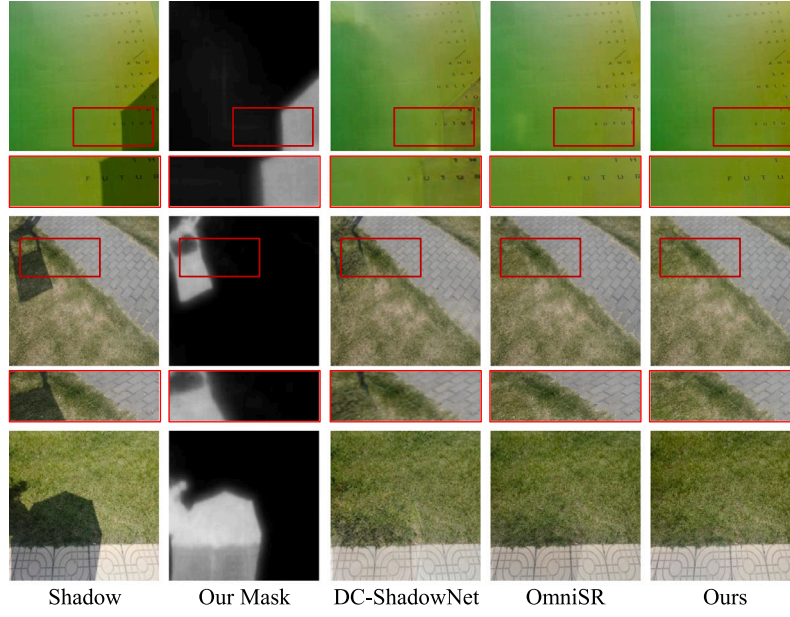
4.1.2. Evaluation metrics

To ensure fairness, we evaluated Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) on the RGB channels and Root Mean Square Error (RMSE) on the LAB color space to assess

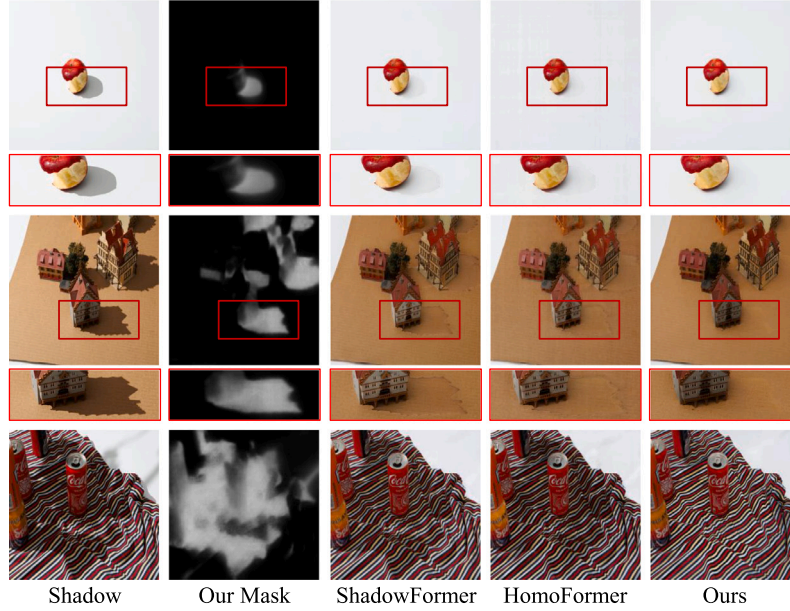
the quality. To quantify the model efficiency, we report both the number of parameters and the FLOPs. Following previous method [4,5,11], all images are resized to 256×256 when evaluating. In addition, given the high resolution of WRSD+ dataset [34,35], we resized all images to 480×360 (1/4 of original) for training and evaluation to facilitate processing and ensure a fair comparison with state-of-the-art methods.

4.1.3. Implement details

We implement our models in PyTorch and train them for 100 and 600 epochs, respectively, using the AdamW optimizer [38] with an initial learning rate of 2×10^{-4} , weight decay 1×10^{-4} , and batch size 7.



(a) Visual comparisons on ISTD dataset [6].



(b) Visual comparisons on WRSD+ dataset [28, 29].

Fig. 5. Visual comparisons of our approach with open-sourced SOTA methods.

Table 4

Quantitative comparisons with SOTA methods on WRSD+ dataset [34,35]. The best and the second best results are **boldfaced** and underlined, respectively.

	Method	PSNR $\uparrow$	SSIM $\uparrow$	RMSE $\downarrow$
480 × 360	Input Image	18.87	0.825	14.76
	UFormer (CVPR'22) [39]	25.68	0.919	6.93
	ShadowFormer (AAAI'23) [4]	25.64	0.918	7.04
	HomoFormer (CVPR'24) [5]	25.87	0.914	7.03
	RASM (MM'24) [11]	25.68	<u>0.920</u>	<u>6.88</u>
	Ours	26.29	0.923	6.63

4.2. Comparison with the state-of-the-art

We presents quantitative comparisons of our model with state-of-the-art (SOTA) methods (DHAN [29], AutoExposure [23], DC-ShadowNet [16], BMNet [24], UFormer [39], ShadowFormer [4], ShadowDiffusion [13], ShadowRefiner [17], FSR-Net [30], HomoFormer [5], RASM [11], Refusion [26], DeS3 [7], OmniSR [8], Diff-Shadow [14], FW-Former [12] and Detail-Preserving [27]) on SRD [25], ISTD+ [6], ISTD [28] and WRSD+ [34,35] datasets with three metrics. Results and complexity are summarized in Tables 1–5. For fair comparison, we primarily use results reported by the original authors. Where results

Table 5

Efficiency comparisons with state-of-the-art (SOTA) methods. The best and the second best results are **boldfaced** and underlined, respectively.

Method	GM	Params (M)↓	GFLOPs↓
DHAN (AAAI'20) [29]	✓	16.4	126.0
AutoExposure (CVPR'21) [23]	✓	19.7	53.0
BMNet (CVPR'22) [24]	✓	0.4	11.6
ShadowFormer (AAAI'23) [4]	✓	11.4	64.6
ShadowDiffusion (CVPR'23) [13]	✓	55.5	728.2
HomoFormer (CVPR'24) [5]	✓	17.8	35.6
RASM (MM'24) [11]	✓	<u>5.2</u>	<u>26.3</u>
FW-Former (JVCi'25) [12]	✓	6.3	27.5
Refusion (CVPRW'23) [26]	✗	131.4	63.4
DeS3 (AAAI'24) [7]	✗	113.7	24.9
OmniSR (AAAI'25) [8]	✗	<u>24.6</u>	78.3
Ours	✗	18.1	<u>36.1</u>

are unavailable, we evaluate provided images or checkpoints using consistent metrics.

Our method achieves state-of-the-art for all metrics on the four datasets. Especially on ISTD+ [6], we achieve superior PSNR in shadow and all image regions, leading the second-best by 2.50 dB and 0.95 dB, respectively, with RMSE reductions of 1.01 and 0.65. And on WRSD+ [34,35], that lack ground-truth masks, those models heavily reliant on ground-truth masks [4,5] face performance bottlenecks. In contrast, our method can generate predicted softmasks entirely without manual annotation, achieving the best performance on WRSD+ dataset [34,35]. Meanwhile, among GM-free methods, our approach achieves the optimal performance in terms of parameter count and the second-best in FLOPs. Compared to diffusion-based models such as Refusion [26] and DeS3 [7], which require massive parameters, and OmniSR [8], which relies on multiple additional priors, our method strikes an optimal balance between computational efficiency and performance. As shown in Figs. 4 and 5, visual comparisons with publicly available methods further confirm the enhanced shadow-removal capability of our model, which benefits from the proposed softmask and structural extractor.

4.3. Ablation study

To validate the contribution of our softmask $\bar{M}_s$ and MLFE, we conduct comprehensive ablation studies on SRD dataset [25]. As summarized in Table 6, we evaluate our baseline model as well as three distinct configurations. Experimental results clearly demonstrate that our softmask and structural module enhance the performance of the baseline model.

4.4. Robustness analysis and limitation

To demonstrate the model's generalization capability, we evaluate our SRD [25] pretrained model on the unseen LRSS [36] and USR dataset [37]. The results are visualized in Fig. 6, where our method successfully predicts softmasks that are well aligned with shadow regions and achieves clean shadow removal.

However, we have to note that under cross-domain generalization with significant color discrepancies, the results may exhibit minor color bias and slight edge discontinuities. Together with the reliance on paired training data, these constitute two limitations of our method.

To address these issues, future work will explore learning from unpaired datasets and develop more effective mechanisms to mitigate color inconsistency and edge artifacts, with the goal of achieving superior generalization and perceptual quality.

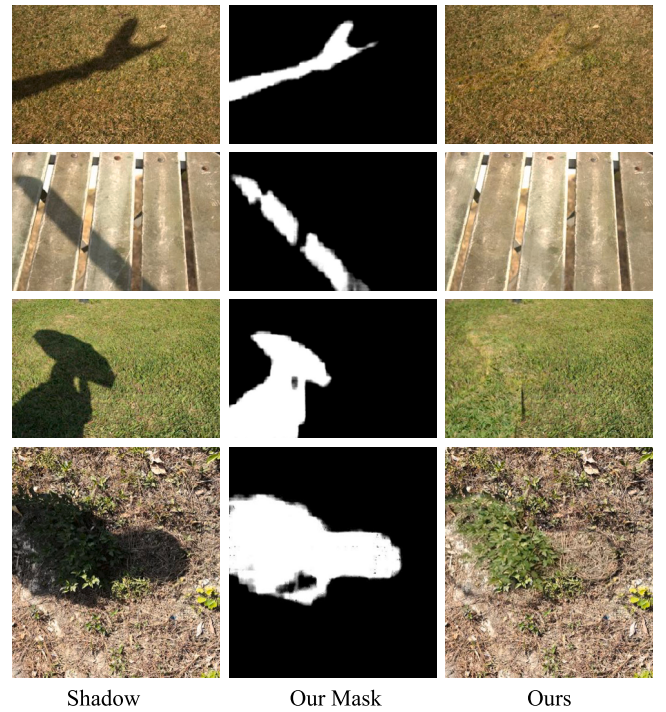


Fig. 6. Visualization on unseen LRSS [36] and USR dataset [37] with SRD [25] pretrained models. The first three rows show successful cases, while the bottom row demonstrates a failure case.

5. Conclusion

In this paper, we presented an illumination-aware softmask guided shadow removal network that eliminates the dependency on ground-truth shadow masks. Our key contribution lies in softmask acquisition, which produces high-quality continuous shadow estimates through multi-scale proportional regression in linear color space. Furthermore, we introduced a MLFE Module that enhances the structural coherence of encoder-decoder layers for better reconstruction performance. Combined with our transformer architecture, the proposed method achieves state-of-the-art performance on multiple benchmarks without ground-truth masks supervision.

CRediT authorship contribution statement

Lianmeng Wei: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sihui Luo:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All the datasets used in this paper are publicly available.

Table 6

Ablation study of different network modules on SRD [25] dataset. PSNR, SSIM and RMSE metrics of the Shadow, Non-shadow, and all regions are reported.

ID	Baseline	MLFE	$\bar{M}_s$	Shadow			Non-shadow			All		
				PSNR↑	SSIM↑	RMSE↓	PSNR↑	SSIM↑	RMSE↓	PSNR↑	SSIM↑	RMSE↓
i	✓			37.56	0.981	6.18	35.53	0.976	3.37	32.63	0.948	3.92
ii	✓	✓		38.09	0.982	5.91	36.20	0.978	3.23	33.30	0.951	3.75
iii	✓		✓	38.87	0.985	5.45	36.94	0.980	2.95	34.01	0.956	3.43
iv	✓	✓	✓	39.00	0.985	5.36	37.08	0.980	2.91	34.14	0.957	3.39

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